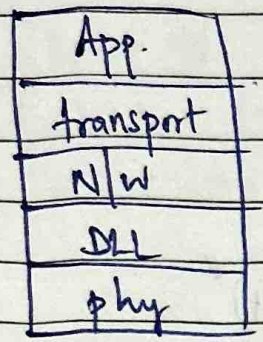
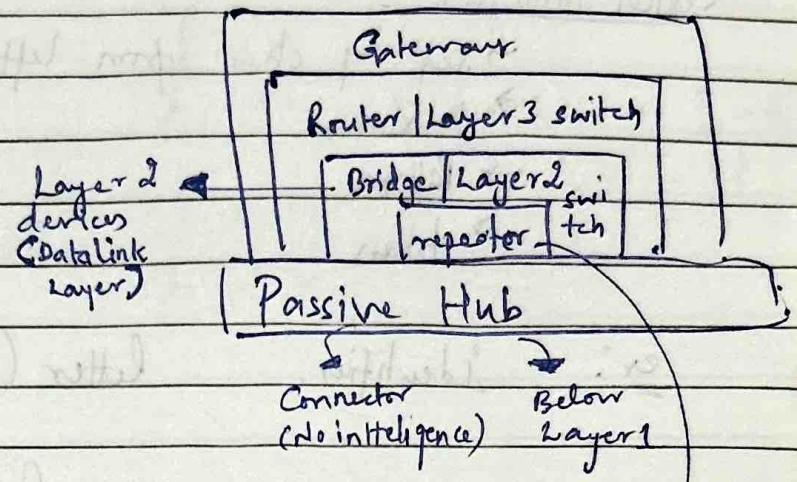
Network Devices

DT. 13 01 2026





Layer 1 [includes Active hub]

OSI:

- Application → providing service to users.
- Present^n → compress/encryption of data packets
- Session → Sync/dialogue control
- transport → process to process & segmentation & reassembly
- N|W → packets
- Data Link → frames
- physical → bits

Active hub: Broadcasting to all connected devices. } Layer 1
repeater: Not an Amplifier but a regenerator.

Bridge

→ filtering capabilities
→ MAC addressing (6 bytes)

Address	Port
MAC	~

port address ↓ 2 bytes

Bridge

Address:

192.168.50.0/24.

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128
64
32
224

Rough

1 0000000

0 1111111

63

110 00000

110 11111

1110

→ Subnetmask: 255:255:255:0

Requirements:

A → 100

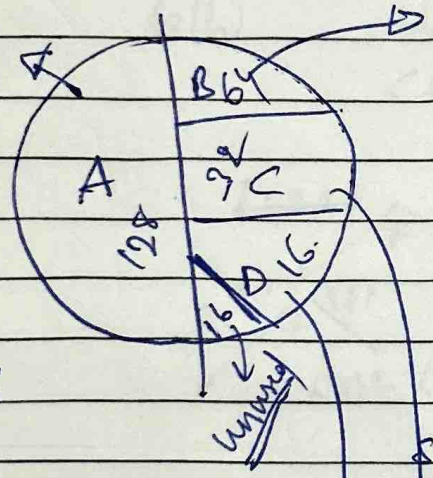
B → 50

C → 20

D → 10 Hosts.

Total Available addresses:

0 to 255 total 256.



~~192.168.50.~~

192.168.50.0

to

192.168.50.127

~~Addresses: 128~~

Hosts: 128-2=126 Hosts

Net ADDr: 192.168.50.0

Broadcast ADDr: 192.168.50.127

Subnet Mask: ~~192.168.50~~ 255:255:255:128

(or)

192.168.50.0/25

192.168.50.128 to

192.168.50.191

total Addresses: 64

Hosts: 64-2 = 62 Hosts.

Net Address: 192.168.50.128

Broadcast Addr: 192.168.50.191

SubNet Mask: 255:255:255:192

(or) 192.168.50.128/26

192.168.50.192 to

192.168.50.223

total Addr: 32

Net Addr: 192:168:50:224 Hosts: 32-2 = 30. Hosts

Broadcast Addr: 192:168:50:239

Hosts: 16-2 = 14

(or)

192.168.50.224/28

Net Address: 192.168.50.224/28

Broadcast Addr: 192.168.50.239

~~Hosts:~~

Net Address: 192.168.50.192

Broadcast Addr: 192.168.50.223

Subnet Mask: 255:255:255:224

(or)

192.168.50.192/27

Spiral

NND

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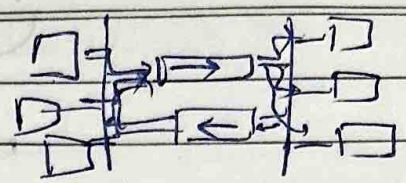
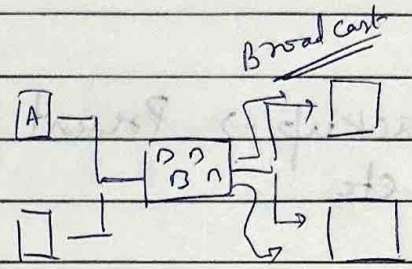
Loops Occur → multiple Bridges
↳ problem with Bridges



Types of Bridges

1. Transparent → devices are unaware of its existence
2. Source Route → Token ring
3. Translational → To convert one n/w format to another

Layered
2. Bridge :-
↳ filtering Capabilities



Src MAC	Port	Dest MAC	Outgoing Port

⇒ Not used much, replaced by Layer 2 switches.

STP (Spanning Tree Protocol)

⇒ Port are blocked to couter Bridge Loop
⇒ Port Blocking.

→ A root Bridge is chosen.
→ In topology, it starts blocking ports of each device
→ Until loops are removed.

Spiral

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Layer 2

Switch

- ⇒ Bridge with multiple ports
- ⇒ Faster performance as compared to Bridges
- ⇒ Filtering Capabilities.
- ⇒ Replaced Bridges in present Scenarios.

- ⇒ Has Buffer Space. to store frames for processing.
- ⇒ Congestion control due to Buffer
- ⇒ Full duplex Communication.

Layer 3

Switch

- ⇒ Transmits data based on IP addr.
- ⇒ Supports L-2 switch functionalities
- ⇒ Dynamic/Static Routing
  - Dynamic → topology is not fixed
  - Static → fixed topologies

	L2 Switch	L3 Switch
Cost	Cost effective	Expensive
Setup & Config	simple	Advanced Configurations
Routing	No Routing, MAC	Support Routing via IP addr.
N/w security	No built in security feat	Have security features ↳ Access control list
QoS	No QoS support	Has n/w prioritization.

Multicasting: Group Multiple devices in a big Network and casting msg to that Grp only.